

PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Introduction to Wavelet Theory

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Introduction and Historical Development	2
1.1	Historical Origin of Wavelets	2
1.2	Different Communities of Wavelet	2
2	Time-Frequency Analysis Limitations	3
2.1	Short-Time Fourier Transform (STFT)	3
2.2	The Wavelet Multi-resolution Approach	3
3	Classification of Continuous and Discrete Wavelet Transforms	4
4	Developments in Wavelet Theory Applications	4

1. Introduction and Historical Development

Definition of a Wavelet

A **wavelet** is a localized, oscillatory wave-like function of effectively finite duration, with a zero mean value. Unlike a sine wave, which extends infinitely in both directions, a wavelet grows and decays over a short period, providing simultaneous **time** and **frequency** localization.

1.1 Historical Origin of Wavelets

The development of wavelet theory is a synthesis of ideas spanning mathematics, physics, and digital signal processing:

- **1910 (Alfréd Haar):** Proposed the first and simplest wavelet, the **Haar wavelet**, as an orthonormal basis for functions. It consists of a step function that changes value abruptly between $+1$ and -1 .
- **1946 (Dennis Gabor):** Developed the Windowed Fourier Transform (or Gabor Transform) to introduce time localization using Gaussian window envelopes.
- **1981 (Jean Morlet):** A geophysicist processing seismic data introduced the concept of the "wavelet of constant shape" with variable scaling and translation to achieve depth resolution.
- **1984 (Alex Grossmann):** Collaborated with Morlet to establish the rigorous mathematical framework for the Continuous Wavelet Transform (CWT) using group theory.
- **1986 (Stéphane Mallat & Yves Meyer):** Formulated **Multi-resolution Analysis (MRA)**, establishing a unified foundation connecting wavelets to subband digital filtering.
- **1988 (Ingrid Daubechies):** Constructed the first family of continuous, compactly supported, orthonormal wavelet bases with maximum vanishing moments (the **Daubechies wavelets**), making wavelets highly practical for computer implementation.

1.2 Different Communities of Wavelet

Wavelet theory is uniquely interdisciplinary, developed independently across multiple scientific domains:

1. **Mathematics Community:** Studied under *Calderón-Zygmund operators*, singular integrals, and harmonic analysis. Focuses on basis representations of Hilbert and Banach function spaces ($L^2(\mathbb{R})$).
2. **Geophysics & Physics Community:** Morlet and Grossmann used wavelets for seismic reflection parsing and quantum mechanics (coherent states, path integrals).
3. **Digital Signal Processing (DSP) & Engineering Community:** Developed wavelets as **filter banks** (subband coding) for audio, image, and video compression.

2. Time-Frequency Analysis Limitations

Traditional signal analysis relies on the **Fourier Transform (FT)**, which maps a time-domain signal $x(t)$ to its frequency spectrum $X(f)$:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

- **Limitation:** FT has *infinite time support* ($e^{-j2\pi ft}$ spans from $-\infty$ to ∞). It yields perfect frequency resolution but completely destroys all time information (we know *what* frequencies exist, but not *when* they occur).

2.1 Short-Time Fourier Transform (STFT)

To introduce time localization, the **Short-Time Fourier Transform (STFT)** multiplies the signal with a sliding window function $w(t)$ centered at time τ :

$$STFT_x(\tau, f) = \int_{-\infty}^{\infty} x(t)w^*(t - \tau)e^{-j2\pi ft} dt$$

- **Limitation:** The window width is **fixed** for all frequencies.
- **Heisenberg's Uncertainty Principle:** The resolutions in time (Δt) and frequency (Δf) are bounded:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

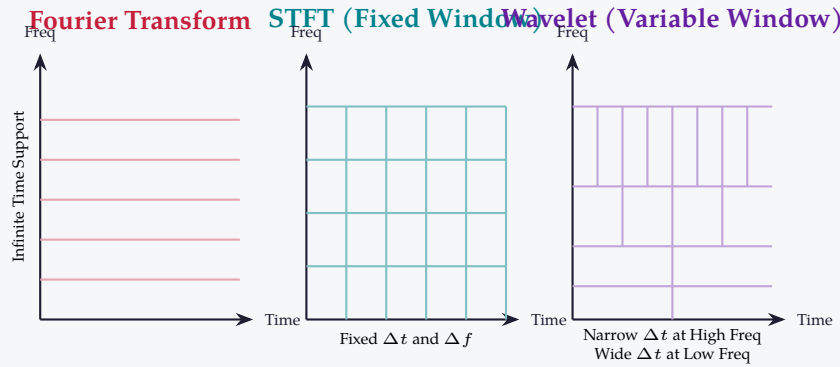
Under STFT, we must choose a fixed window. A narrow window gives good time resolution but poor frequency resolution (Wideband STFT). A wide window gives good frequency resolution but poor time resolution (Narrowband STFT).

2.2 The Wavelet Multi-resolution Approach

The Continuous Wavelet Transform (CWT) resolves this restriction by using a **variable-width window**:

- At **high frequencies**, it uses narrow windows (high time resolution, lower frequency resolution) to capture rapid transients and edges.
- At **low frequencies**, it uses wide windows (high frequency resolution, lower time resolution) to capture slow-varying stationary components.

Time-Frequency Tiling Grids Comparison



3. Classification of Continuous and Discrete Wavelet Transforms

Wavelet transforms are classified based on how the scaling parameter a and translation parameter b are sampled:

- 1. Continuous Wavelet Transform (CWT):** The parameters a and b vary continuously over $\mathbb{R} \setminus \{0\}$ and $\mathbb{R}$ respectively. This maps a 1D time-domain signal to a 2D time-scale plane, generating a highly redundant representation. It is ideal for signal analysis, singularity detection, and scientific research.
- 2. Discrete Wavelet Transform (DWT):** The parameters a and b are sampled discretely. Typically, a **dyadic grid** is used:

$$a = 2^j, \quad b = k2^j, \quad j, k \in \mathbb{Z}$$

This eliminates representation redundancy, allowing for an orthogonal basis decomposition. It is highly efficient for data compression (e.g. JPEG 2000), denoising, and digital hardware implementation.

Feature	Continuous Wavelet (CWT)	Discrete Wavelet (DWT)
Scaling (a) & Shift (b)	Continuously variable ($a, b \in \mathbb{R}$).	Discretely sampled ($a = 2^j, b = k2^j$).
Redundancy	High (infinite overlapping coefficients).	None (orthogonal / non-redundant).
Reconstruction	Uses stable double integral.	Uses digital filter banks (perfect reconstruction).
Best Use Case	Singularity parsing, physics, and analysis.	Signal compression, denoising, and coding.

4. Developments in Wavelet Theory Applications

Since its formalization, wavelet theory has seen rapid developments and applications across various engineering and scientific fields:

- **Seismic Signal Processing:** Jean Morlet's initial development for processing geophysics reflection waves to localize underground oil deposits.
- **Data & Image Compression:** Standardized in JPEG 2000 (DWT) and FBI fingerprint databases, allowing high-quality storage with minimal memory.
- **Biomedical Engineering:** Analyzing ECG (electrocardiogram) and EEG (electroencephalogram) signals for transient spike detection (such as epilepsy seizures) and denoising.
- **Communication Systems:** Subcarrier modulation schemes like Discrete Wavelet Multi-Tone (DWMT) to suppress inter-carrier interference in multicarrier systems.

Quick Revision — Module 1

Introduction to Wavelet Theory | PE-EC703C

Key Time-Frequency Equations

Fourier Transform	$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$
Short-Time FT (STFT)	$STFT_x(\tau, f) = \int_{-\infty}^{\infty} x(t)w^*(t - \tau)e^{-j2\pi ft} dt$
Heisenberg Uncertainty	$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$ (Limits simultaneous time-frequency resolution)

One-Line Definitions

- **Wavelet:** A localized, oscillatory wave-like function of finite duration with a zero mean value.
- **Fourier Transform:** Spectral mapping with infinite time support, providing no temporal location details.
- **STFT:** Time-frequency mapping that uses a fixed window width, imposing rigid time-frequency resolutions.
- **Multi-resolution Approach:** Analysis method that analyzes signals with narrow windows at high frequencies and wide windows at low frequencies.
- **CWT:** Wavelet transform where parameters vary continuously, providing a redundant, high-resolution representation.
- **DWT:** Wavelet transform where parameters are sampled on a discrete grid, providing an orthogonal representation.

Frequently Asked / Viva Questions

Q1. Why can the Fourier Transform not analyze non-stationary signals?

The Fourier Transform integrates over all time ($-\infty$ to ∞), meaning it assumes the signal's spectral components do not change over time. It has no time localization, so it cannot identify *when* a transient event or change in frequency occurs.

Q2. How does the Short-Time Fourier Transform (STFT) attempt to fix the limitations of the Fourier Transform?

The STFT divides the signal into short segments by multiplying it with a sliding window function. The Fourier Transform is then computed for each segment, providing a time-frequency spectrogram.

Q3. What is the drawback of the STFT?

The STFT uses a fixed window size for all frequencies. Due to Heisenberg's Uncertainty Principle ($\Delta t \cdot \Delta f \geq 1/4\pi$), a narrow window gives good time

Frequently Asked / Viva Questions (contd.)

resolution but poor frequency resolution, while a wide window does the opposite.

Q4. How do Wavelets resolve the resolution trade-off of the STFT?

Wavelets use variable-width windows: narrow windows (small scales) at high frequencies for capturing rapid transitions, and wide windows (large scales) at low frequencies for capturing steady long-term behaviors.

Q5. What is the main difference between CWT and DWT in terms of redundancy and representation?

CWT translates and scales the mother wavelet continuously, leading to high redundancy and overlapping information, making it best for signal analysis. DWT samples parameters on a discrete dyadic grid, yielding a non-redundant orthogonal representation that is highly efficient for data compression.

Time-Frequency Analysis Features

Feature	Fourier Transform (FT)	Short-Time FT (STFT)	Wavelet Transform
Time Resolution	None (integrates over all time).	Fixed (determined by window width).	Variable (excellent at high frequencies).
Freq Resolution	Perfect (at all frequencies).	Fixed (determined by window width).	Variable (excellent at low frequencies).
Window Shape	Infinite ($e^{-j2\pi ft}$).	Fixed envelope ($w(t)$).	Scaled mother wavelet ($\psi(t/a)$).
Best Use Case	Stationary signals.	Quasi-stationary signals.	Non-stationary/transient signals.